INTRODUCTION

This procedure is a guide to help Field Engineers when assembling the Lube Oil Piping on an MS6001B gas turbine split base unit. This piping is Material Shipped Direct (MSD) from the vendor to the site.

REFERENCE DOCUMENTS

No.	Description
ML 0969	Piping Arrangement, Base Interconnect
ML 0969	Piping Arrangement, Interconnect – Lube Oil
ML 0969	Piping Arrangement, Interconnect – Fuel Gas
ML 0905	Piping Arrangement, Lube Oil Feed and Drain – Base
ML 0906	Piping Arrangement, Lube Oil Feed and Drain – Load

LUBE OIL PIPING ASSEMBLY PROCEDURE

The following procedure is a proven method of installing and assembling the 18 inches square Lube Oil drain manifold located on the left side of the accessory and turbine bases.

1. Inventory MSD crates marked ML 0969 and ML 0906 when delivered to site. This is the Customer's responsibility. Store parts in a clean, dry location.

2. Report shortages or damaged parts to appropriate product department

3. Rig and move the 4 segments of the 18 inch square drain pipe (Figure 1) and set down along the left side of the turbine and accessory bases.

4. Set Segment #3 of the drain pipe in position next to the turbine base on wood blocking. It must be set on the blocking because the two IGV tubing connections coming out of Junction Box #2 must line up with the two IGV tubing connections coming out of the turbine base side wall. See Figures 2 and 6.

   Note: Assemble the 4 inch dresser coupling on the pipe coming out of the turbine base **before** setting the square drain pipe in position. This must be done in order to attach the coupling (see Figure 6).

5. Segment #4, which includes Junction Box #3, should be set on wood blocking downstream of segment #3. Apply a thin coat of RTV on one side of the square 18 inch bolting flange gasket, and fit on Segment #3 flange (see Figure 3).

6. Make up the square 4 inch lube oil pipe flange that runs inside of the 18 inches square drain pipe. Assemble the O-ring gasket **before** making up the flange, and lubricate the O-ring with a small amount of petroleum jelly. This pipe should be labeled "OR" (Lube Oil). See Figure 3.

7. Apply a thin coat of RTV to the front side of the 18 inch square drain pipe bolting flange gasket, and make up all of the bolts (see Figure 3).

8. The next piece of 18 inch square drain pipe to be assembled is Segment #2. This segment has one bolting flange that connects to Segment #3. The other end has a slip joint that slides into Junction Box #1 and is welded after Junction Box #1 has been assembled to the accessory base. Apply a thin coat of RTV to both sides of the 18 inch square drain pipe bolting flange gasket before making up the bolts (see Figure 4).

9. Move Segment #1 (Junction Box #1) next to the accessory base, leaving a gap of about 8 to 10 inches. It is easier to assemble the accessory base side of flex hoses OH, OLT, OLT-2, and OR **before** bolting Junction Box #1 to the accessory base. Apply a thin coat of RTV on the back side of the square bolting flange gasket that fits on the accessory base and then set in place. Slip the 4 flex hoses through the top of Junction Box #1 and make up bolting flanges. Assemble the O-ring gaskets **before** making up the flanges and lubricate with a small amount of petroleum jelly (see Figures 5 and 8).

10. With the flex hoses assembled on the Accessory base side, apply a thin coat of RTV to the front side of the square gasket. Move Junction Box #1 to the bolting flange and assemble the bolts (see Figure 8).

11. Assemble flex hoses OH, OLT, OLT-2, and OR to their corresponding pipe runs in the 18 inch square drain pipe. Assemble the O-ring gaskets **before** making up the flanges and lubricate with a small amount of petroleum jelly (see Figure 8).

    Note: Do not weld the slip joint between Sections #1 and #2 until the header assembly has been completed.

12. Fit and assemble the lube oil feed and drain flex hoses at the aft end of the turbine base (left side). These 2 hoses (PT. #20 and #21) fit through an opening in the side wall of the turbine base. See Figure 7.

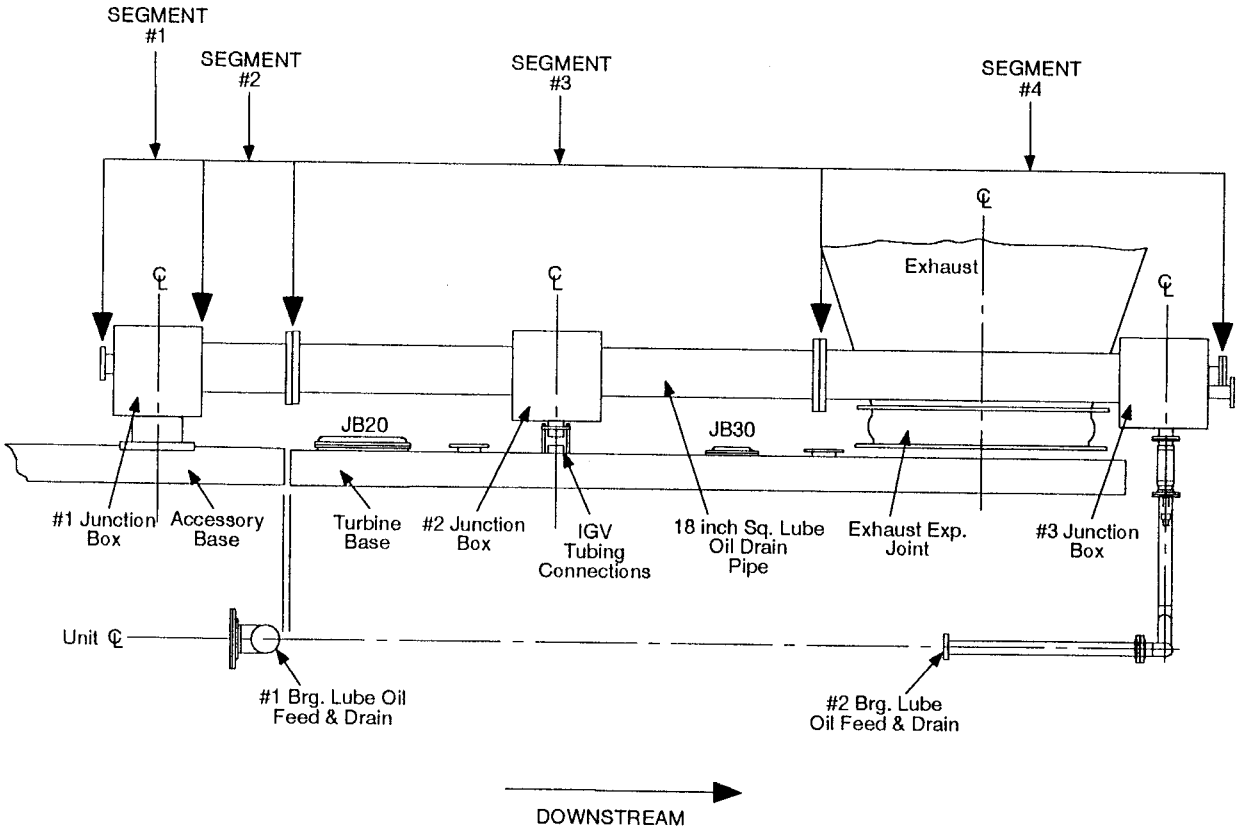


Figure 1. Drain Pipe – Segments #1, #2, #3 and #4

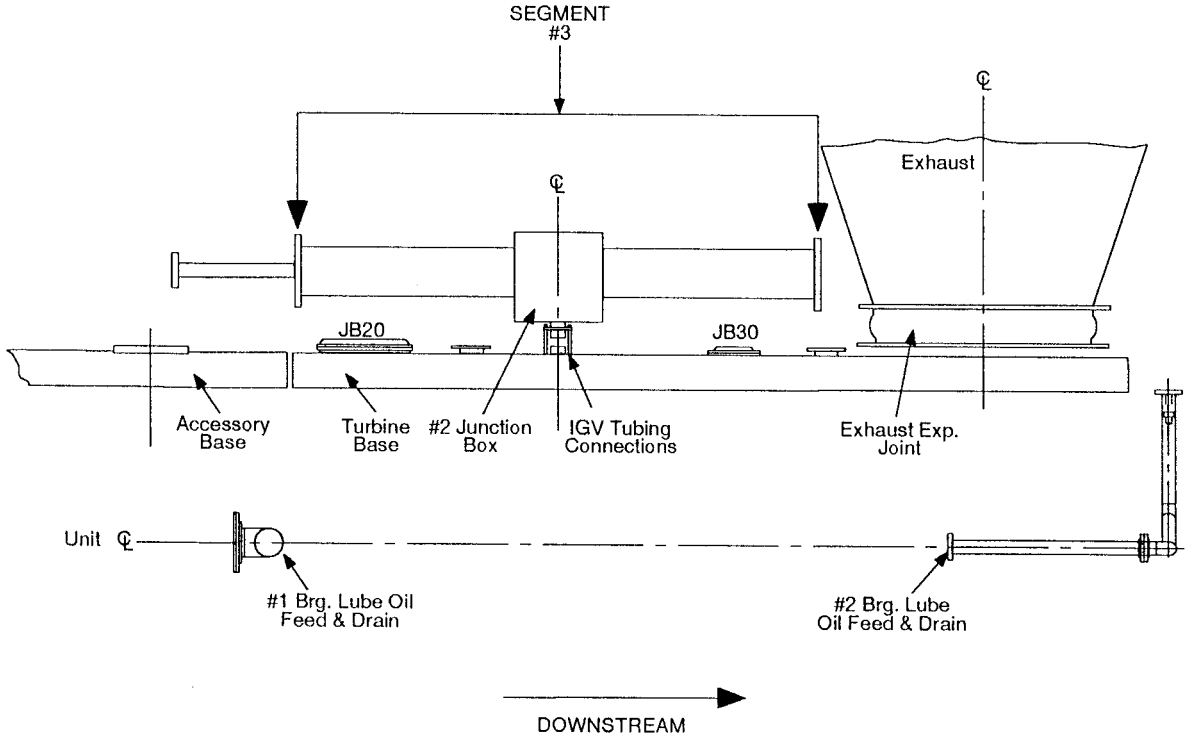


Figure 2. Drain Pipe – Segment #3

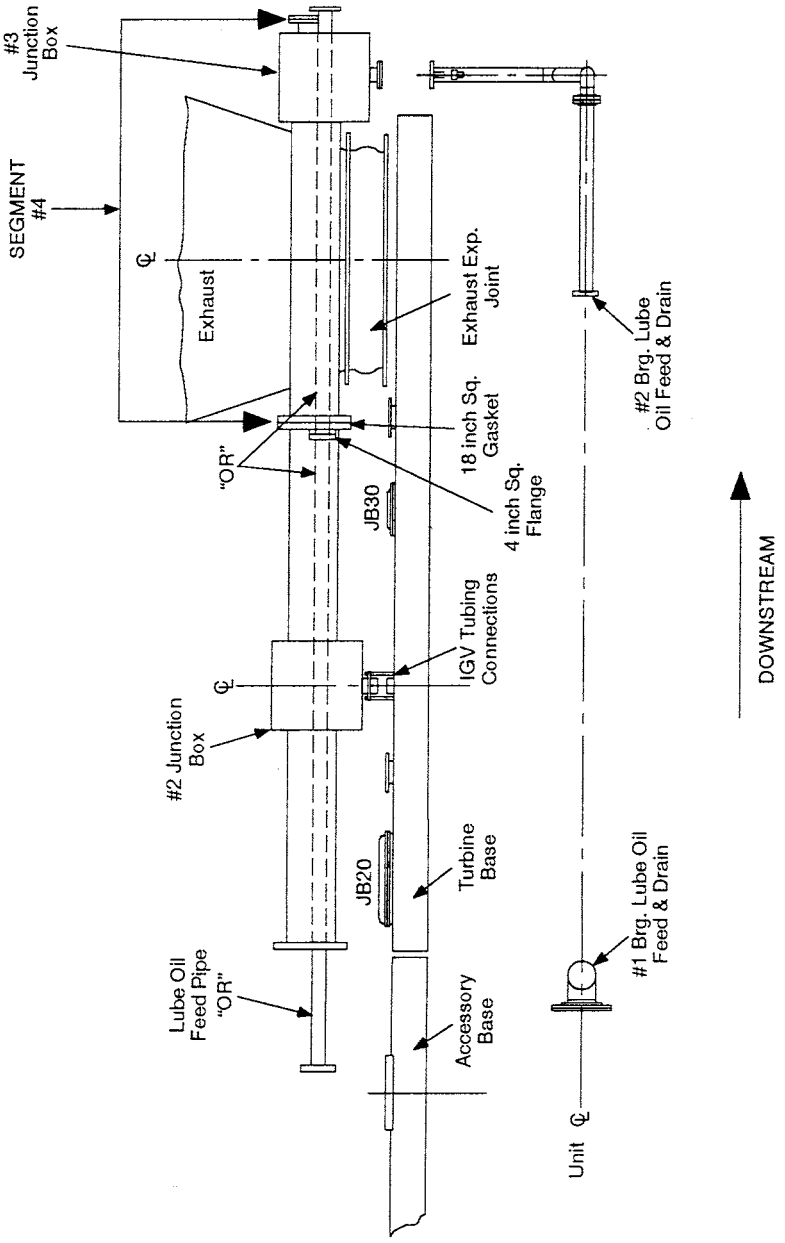


Figure 3. Junction Box #3 Set on Drain Pipe Segment #4

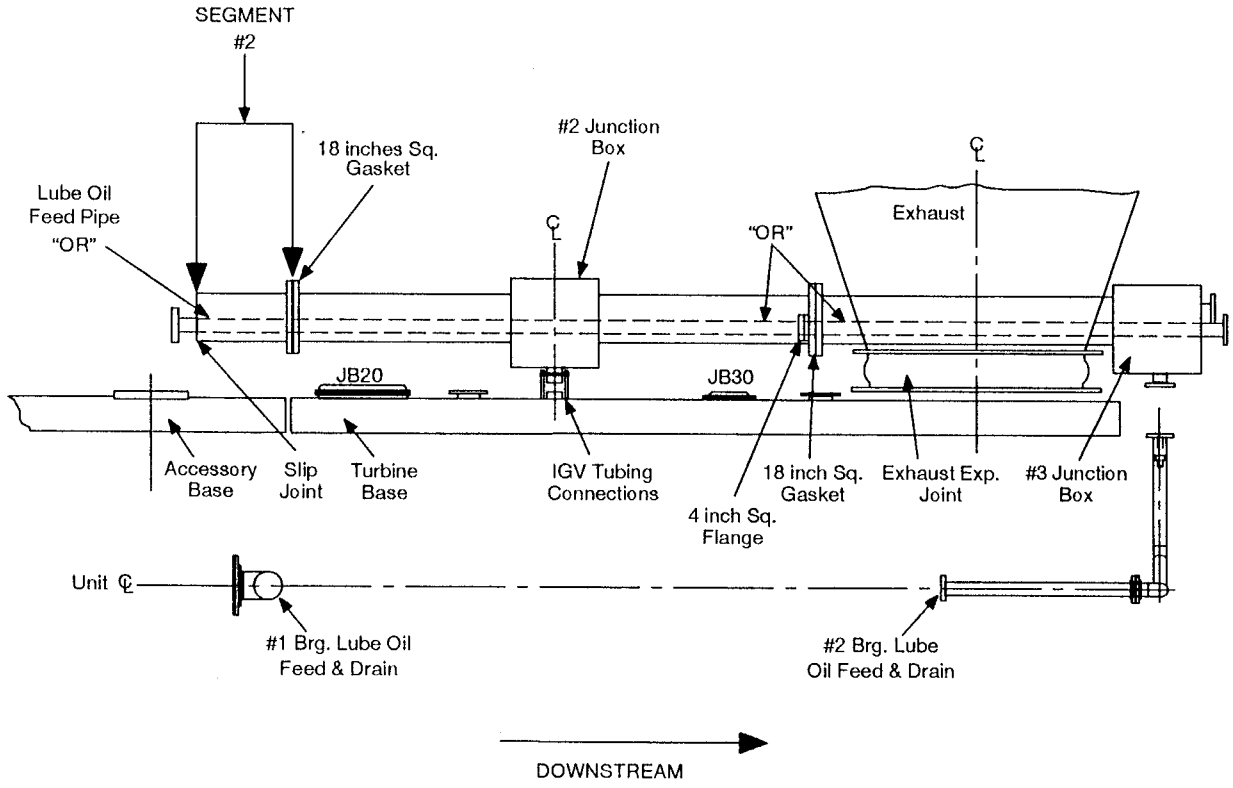


Figure 4. Assembly – Drain Pipe Segment #2

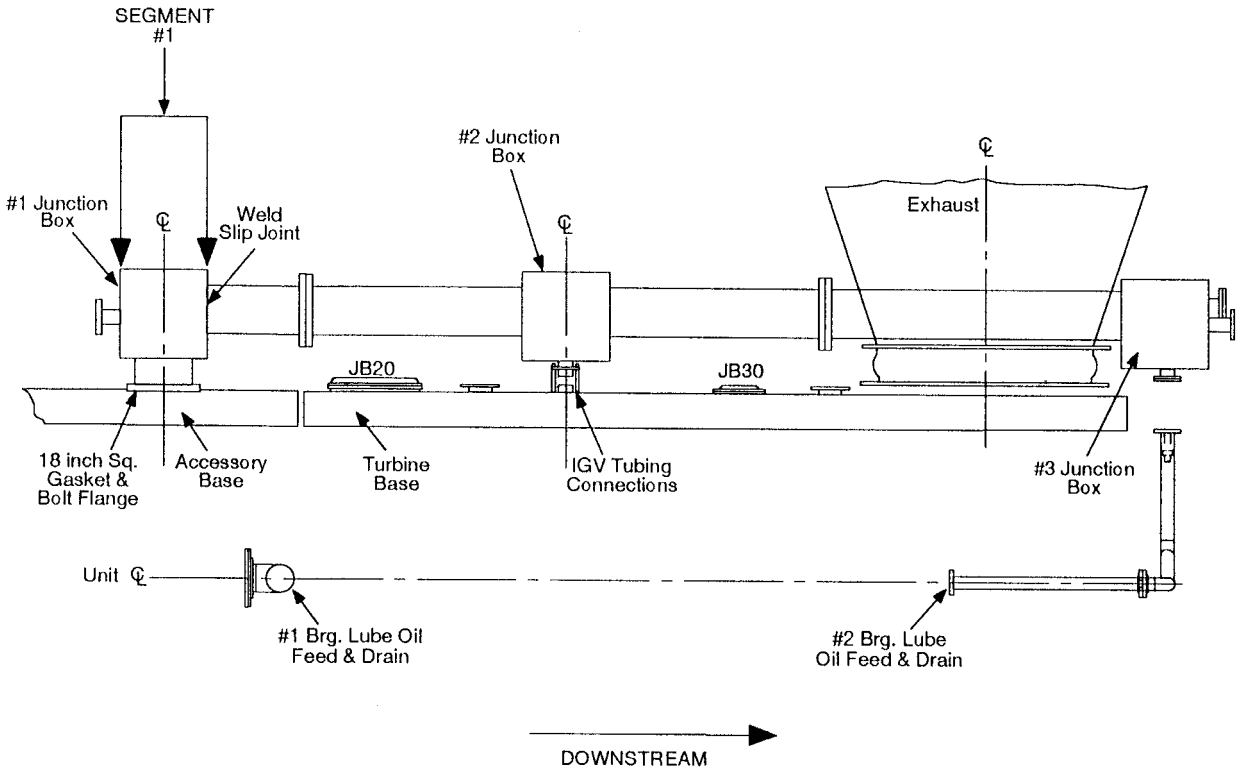


Figure 5. Assembly – Drain Pipe Segment #1

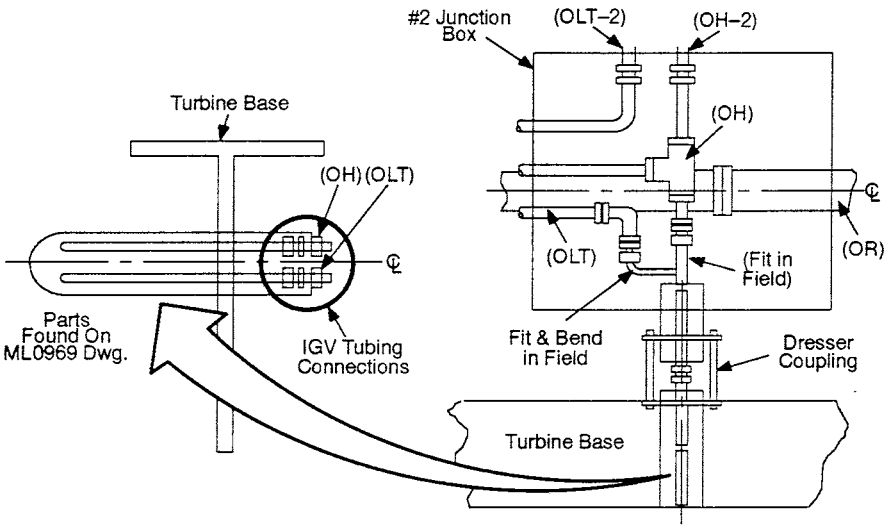


Figure 6. Connecting Tubing Lines OH and OLT in Junction Box #2

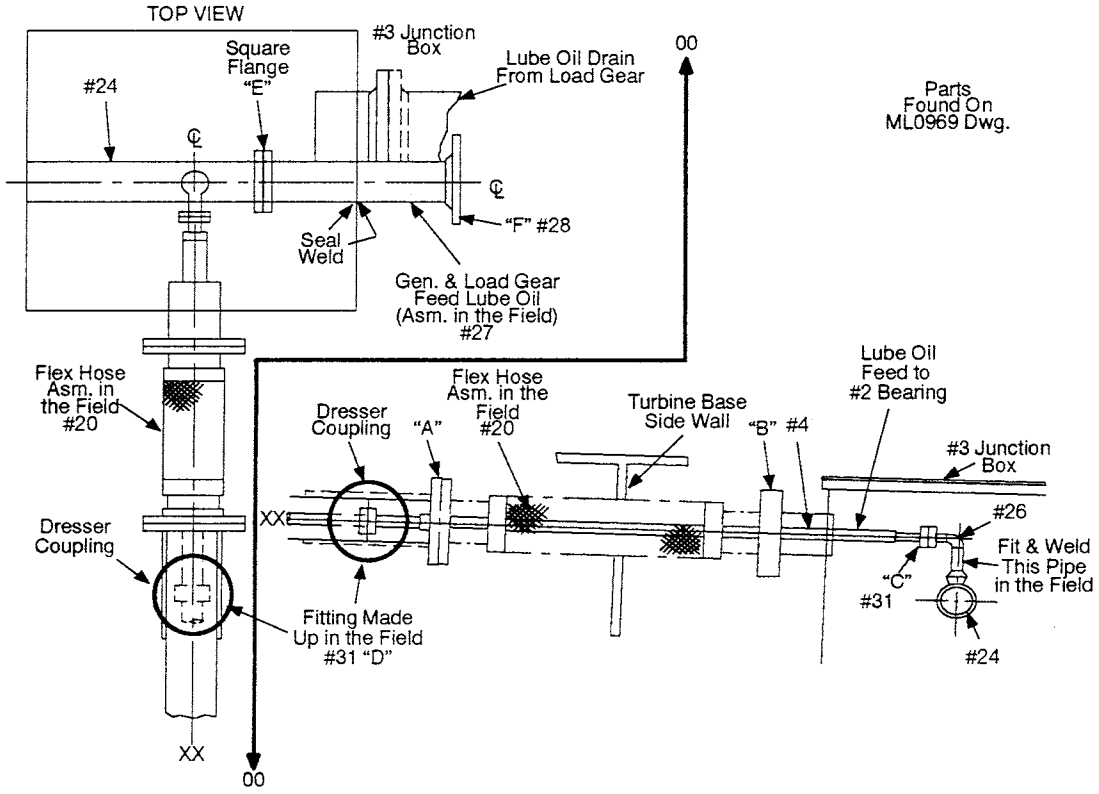


Figure 7. Assembling Lube Oil Feed and Drain Flex Hoses – Turbine Base Aft End

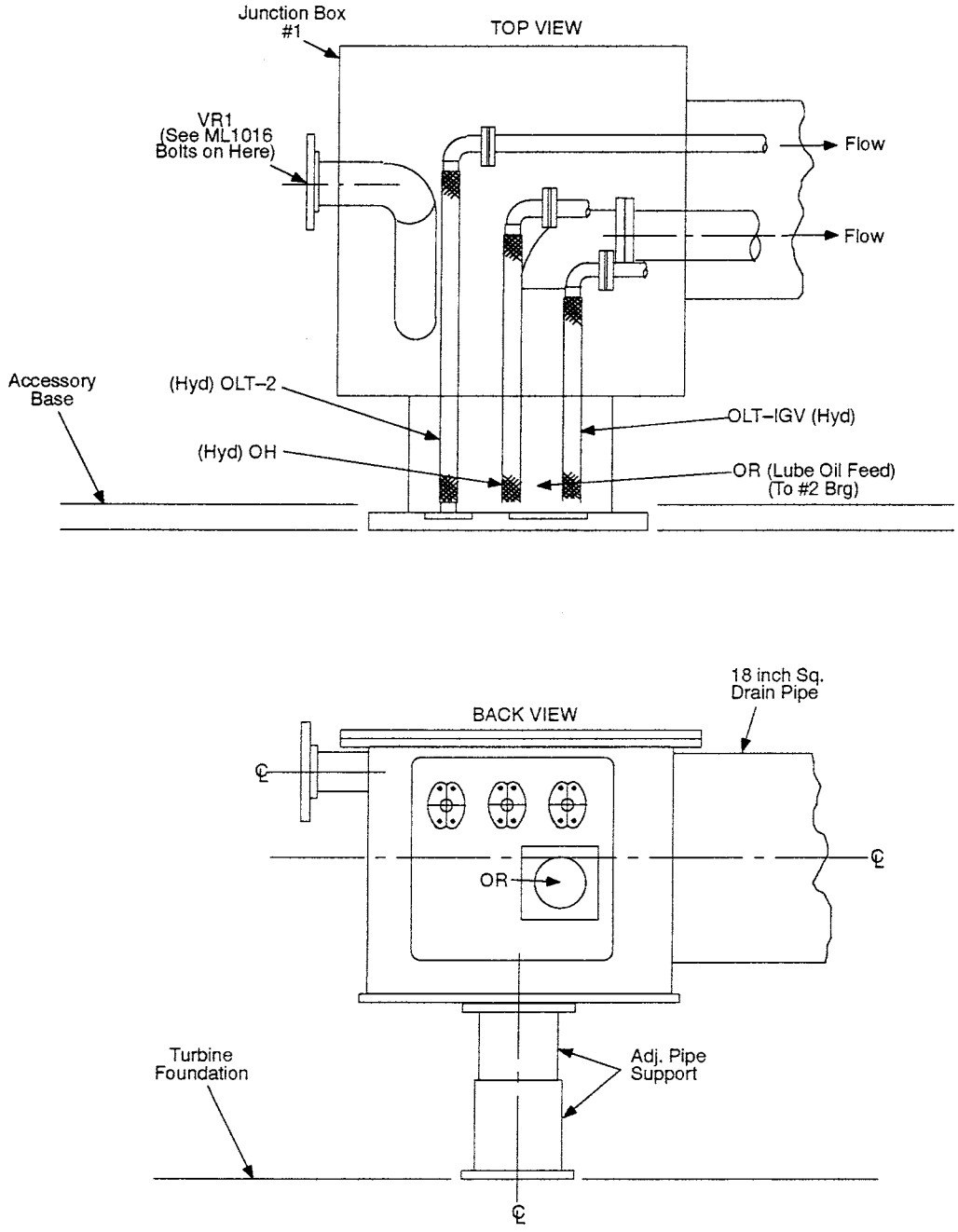


Figure 8. Assembling Junction Box #1 to Bolting Flange

13. Bolt the #2 bearing drain hose to a flange on the back side of Junction Box #3.

14. Slide Flex Hose #21 in through Flex Hose #20. Next, slide Flex Hose #21 in through pipe connection "B" welded in the turbine side wall of Junction Box #3 (see Figure 7). Make up PT. #31 at "C" flange. Assemble the O-ring gasket first **before** making up the flange, and lubricate with petroleum jelly. **Do not weld** PT. #26 to PT. #24 at this point in in the procedure.

    1. Disassemble the dresser coupling (Figure 7) and make up the "D" flange (PT. #31). Assemble the O-ring gasket first **before** making up the flange. Make up flanges "A" and "B" and then reassemble the dresser coupling (reference drawings ML 0905 and ML 0969 for parts).

    2. PT. #26 should now be fitted and welded to PT. #24 (Figure 7).

15. In order to connect the two hydraulic tubing lines labeled "OH" and "OLT" (located inside Junction Box #2) with the two IGV tubing lines coming out of the turbine base side wall, you must first obtain two extra ¾-inch Swedglock tubing fittings. These fittings are needed to make up tubing run "OLT." An extra ¾-inch tubing nut is needed to connect this run on the turbine side (see Figure 6).

16. The "OH" tubing run takes a straight piece that must be custom fit in the field. An extra ¾-inch tubing nut is needed to connect this run on the turbine side (see Figure 6).

17. Slip pipe #27 through the hole in the back wall of Junction Box #3. Assemble the O-ring gaskets **before** making up the square flanges at ("E") between pipe #24 and pipe #27, and lubricate with a small amount of petroleum jelly (see Figure 7).

    1. Now fit slip-flange "F" to pipe #27 and seal weld (see Figure 7).

    2. After pipes #24 and #27 have been bolted together and put in the correct position, seal weld pipe #27 to the back side wall of Junction Box #3 (see ML 0969 Piping Arrangement, Base Interconnect - Note 9).

18. The following method has been found to be the easiest way of assembling ML 0906 (the lube oil feed and drain piping running from Junction Box #3 to the generator and load gear).

    1. The lube oil used in the alignment phase of the installation should have already been removed from the load gear.

2. Remove the 16-inch flange cover at point "A" (see Figure 9) (left side of load gear), and clean the inside cavity of the gear to remove any oil or dirt.

3. Assemble Flex Hose PT. #4 to gear at point "A" and make up bolts.

4. Fit pipe PT. #8 in between point "B" and point "C" and make adjustments at point "C."

5. Seal weld the slip-flange at point "C" to pipe PT. #8 and make up bolts at points "B" and "C." Insert gaskets in flanges **before** making up bolts (see Figure 9).

6. Remove flange cover from gear at point "D" (Figure 9). The flange in the gear is recessed and takes a Garlock gasket, which may have to made on site.

7. Insert gasket and assemble pipe PT. #6 to the load gear at point "D" and make up bolts.

8. Insert gasket and assemble pipe PT. #5 to pipe PT. #6 at point "E" and make up bolts.

9. Insert gasket and assemble flex hose PT. #2 to pipe PT. #5 at point "F" and make up bolts.

10. Fit pipe PT. #9 between points "F" and "H" and make adjustments at point "H" end of the pipe.

11. Seal weld the slip-flange at point "H" to pipe PT. #9 and make up bolts at points "G" and "H." Insert gaskets in flanges **before** making up bolts (see Figure 9).

12. Insert gasket at point "K" and assemble flex hose PT. #1 to the generator and make up bolts.

13. Fit pipe PT. #5 to flex hose PT. #1 and make adjustments at point "J" end of the pipe.

14. Seal weld the slip-flange at point "J" to pipe PT. #5, insert gasket and make up bolts at points "J."

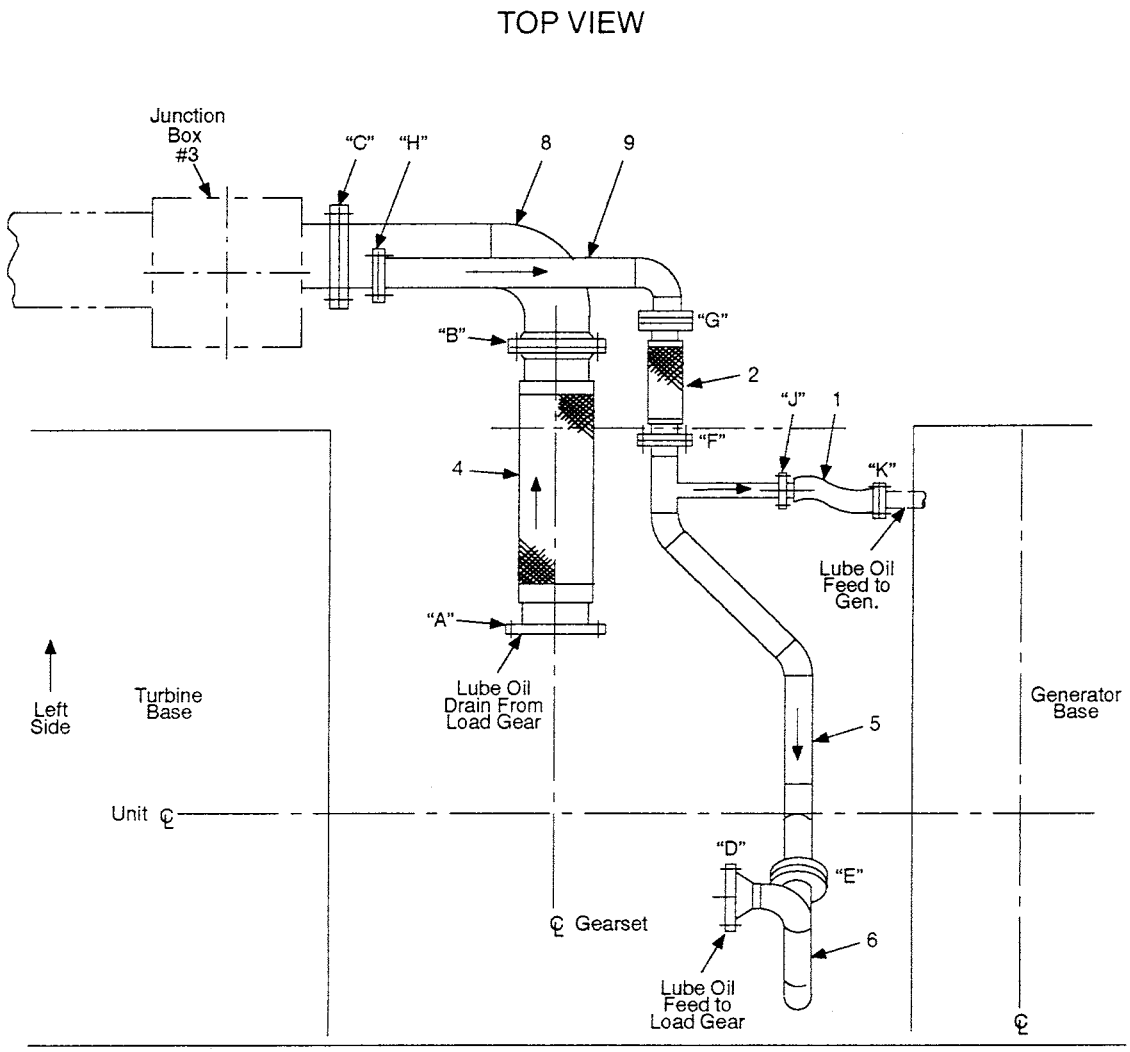


Figure 9. Lube Oil Feed and Drain Piping Assembly

19. Remove the 11-inch flange cover at point "L" (see Figure 10) (right side of load gear) and clean the inside cavity of the gear to remove any oil or dirt.

20. Insert gasket at point "L" and assemble pipe PT. #7 to the load gear and make up bolts (see Figure 10).

21. Insert gaskets at points "M" and "N." Assemble flex hose PT. #3 and make up bolts (see Figure 10).

22. Remove the shipping cover from lube oil pipe flange at point "P" on the left side of the accessory base (see Figure 11).

    1. Clean flange free of any dirt and old oil.

    2. Insert gasket at point "P." Assemble flex hose PT. #35 and make up bolts.

    3. Remove shipping cover from lube oil pipe flange at point "R" on the front of Junction Box #1 (see Figure 11).

    4. Clean flange free of any dirt and old oil.

    5. Insert gasket at point "R." Assemble VR-1 valve and make up bolts.

    6. Insert gasket at point "Q." Assemble flex hose PT. #35 to VR-1 valve and make up bolts.

    7. Cut and fit tubing PT. #33, and assemble between VR-1 valve and Junction Box #1 using PTS. #9 and #5007.

23. Assemble the turbine #1 bearing lube oil feed and drain piping in the following manner.

    1. Remove shipping cover at point "S" (see Figure 12) located on the turbine side of the accessory base.

    2. Clean flange free of any dirt and old oil.

    3. Slip pipe PT. #16 in through pipe PT. #15. Insert gasket at point "S" and insert gasket PT. #29 at point "T." Make up pipe flange.

    4. Assemble PT. #30 Dresser Coupling down over pipe PT. #15 and slide it down enough so that point "U" is exposed.

    5. Fit pipe PT. #13 between points "U" and "W."

6. Slip pipe PT. #13 in through pipe PT. #14, insert gasket PT. #34 at point "V" and make up flange (see Figure 12).

7. Disassemble pipe slip-flange at point "W" and slide it back over pipe PT #14.

8. Assemble pipe PT. #14 into the housing marked ML 0905 on Figure 12, slide slip-flange back in place and make up bolts.

9. Insert gasket PT. #28 at point "U" and make up bolts.

10. Slide PT. #30 Dresser Coupling up over pipe PT. #14 so that it is centered half way between pipe PTS. #14 and #15, then make up dresser coupling bolts.

11. Position the two PTS. #8 and weld to pipe PT. #14 and PT. #15.

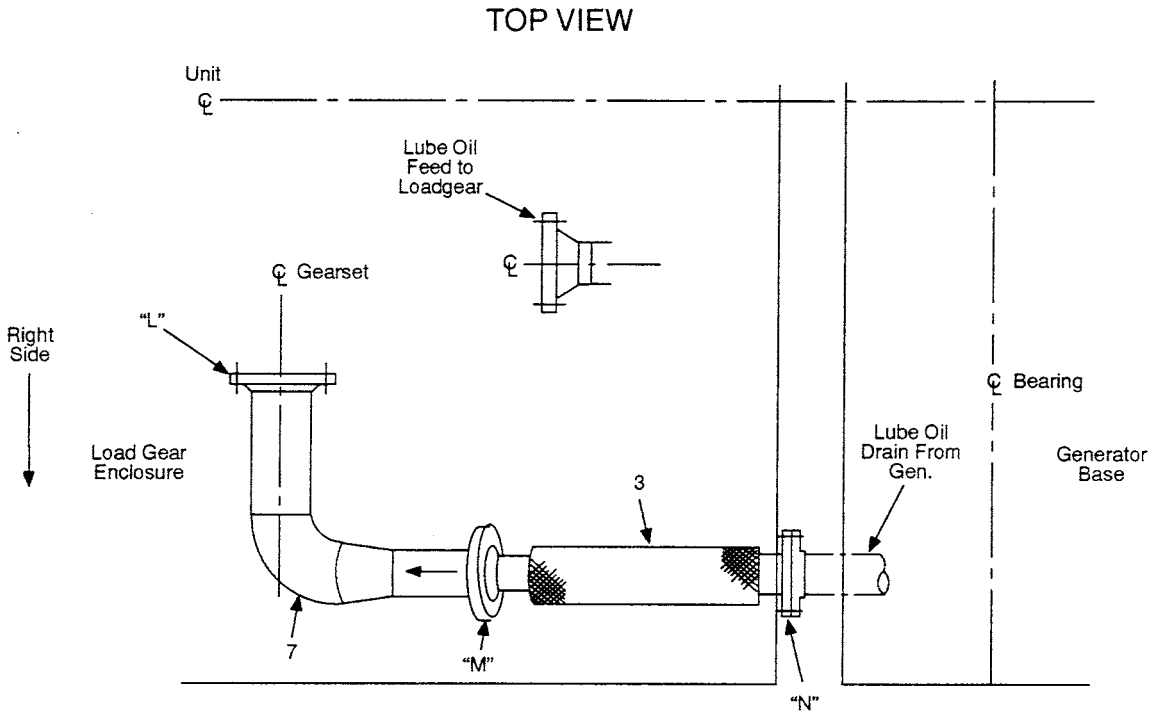


Figure 10. Assembling Pipe to Load Gear

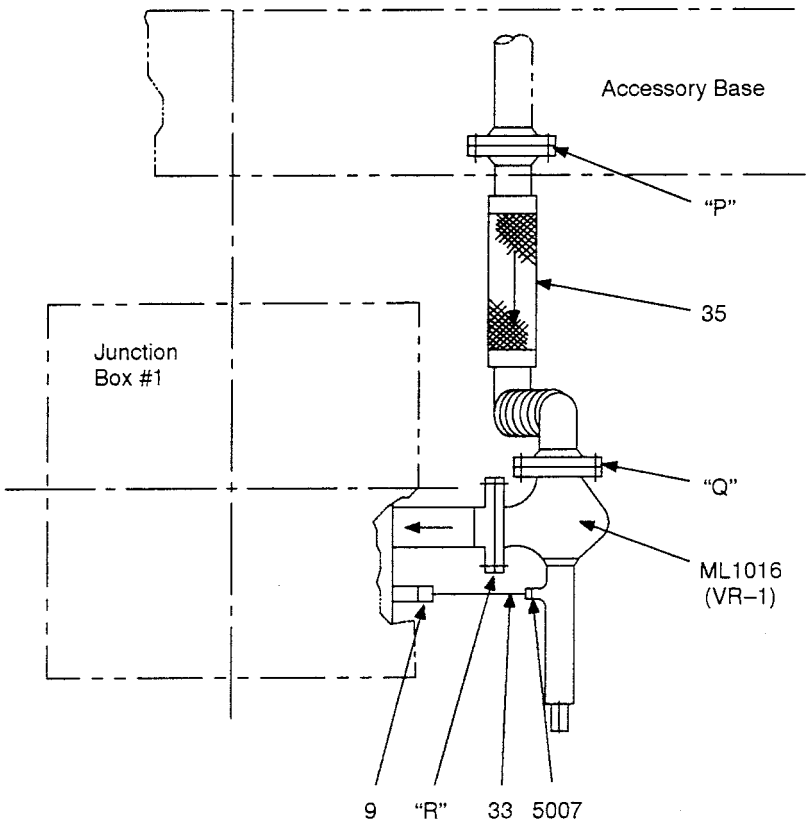


Figure 11. Assembling VR-1 Valve to Junction Box #1

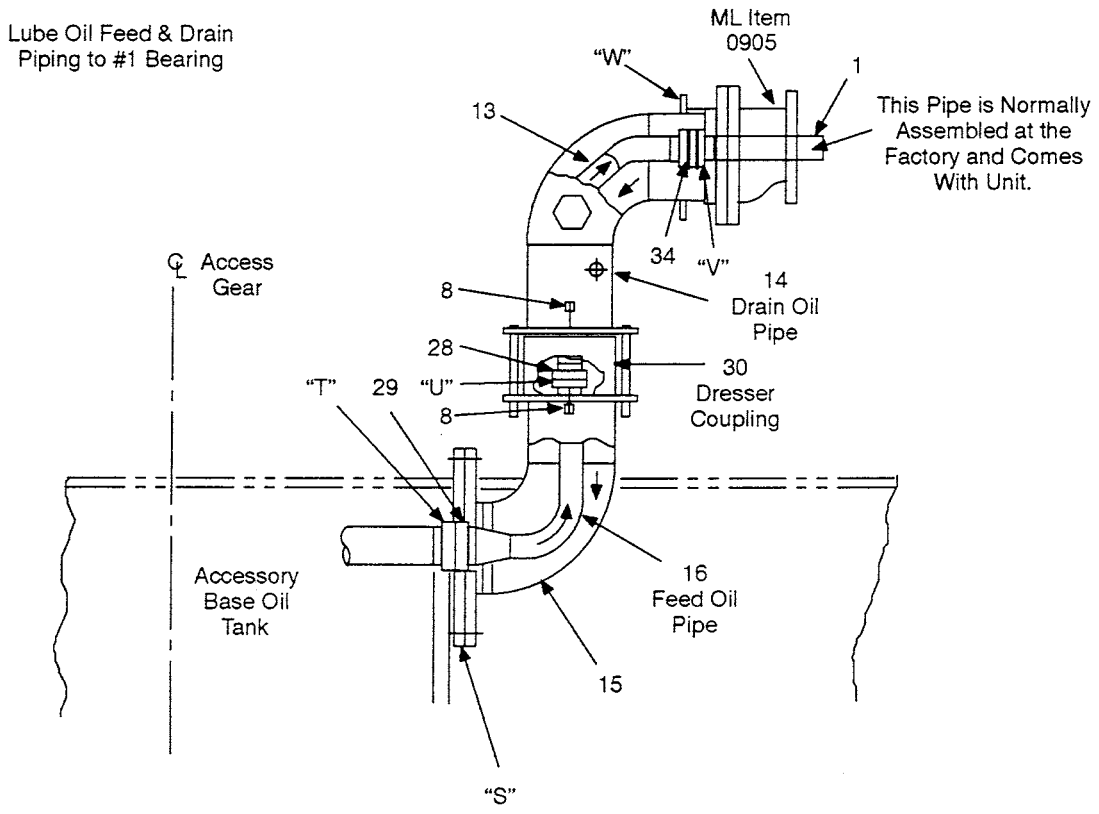


Figure 12. Assembling Turbine #1 Bearing Lube Oil and Feed Drain Piping

24. Field assembly of the hydraulic pipe and fuel gas pipe running from the gas module to the turbine should be performed in the following manner. See Figure 13.

    Note: The fuel gas module should have already been set in its proper location.

    1. Remove shipping cover at point "X" located on the left side of the turbine base.
    2. Clean flange free of any dirt and old oil.
    3. Insert two gaskets (PT. #3435) one on each side of PT. #6 strainer and assemble in the left side of the turbine base at point "X" on Figure 13.
    4. Assemble flex hose PT. #5 by sliding it down under the 18-inch square lube oil drain pipe manifold and make up bolts at point "X."
    5. Remove shipping cover at point "Y" located on the upstream side of the gas module.
    6. Clean flange free of any dirt and old oil.
    7. Insert gasket PT. #3435 at point "Y" pipe flange and make up bolts.
    8. Remove shipping covers at points "Z," "AA," "BB," "CC," "DD," and "EE."
    9. Clean flange free of any dirt and old oil.
    10. Insert O-rings in PT. #3 pipe flanges at points "Z" and "AA" and make up bolting (hydraulic pipe drain).
    11. Insert O-rings in PT. #1 pipe flanges at points "DD" and "EE" and make up bolting (hydraulic pipe "OH"-2).
    12. Insert O-rings in PT. #2 pipe flanges at points "BB" and "CC" and make up bolting (hydraulic pipe "OLT"-2).

25. Assemble the turbine Interconnecting Lube Oil header (also known as the hair pin pipe) in the following manner.

    Note: Parts #1, 2, 3, 10, 0224, 1043, 1023, 32, 33, and 5010 can be found on GE dwg. 103E8107g001, which is part of ML 0969.

    1. Assemble both PTS. #10 (pipe supports) to the left lower side of the accessory base (see Figure 14).

2. Remove shipping cover at point "A" (see Figure 14) located on the left side of the accessory base.

3. Clean flange free of any dirt and old oil.

4. Insert gasket and assemble flex hose PT. #1 at point "A" and make up bolts.

Note: Parts nos. 3, 4, 5, and 6 can be found on GE dwg. 213C1284g001, which is part of ML 0969.

5. Insert gasket at point "B," assemble pipe PT. #5 and make up bolts.

6. Insert two gaskets, one on each side of Orifice ML 1043 at point "C," assemble pipe PT. #3, and make up bolts.

7. Remove shipping cover at point "E" (see Figure 14) located on the left side of the accessory base.

8. Clean flange free of any dirt and old oil.

9. Insert gasket at point "E," assemble flex hose PT. #2, and make up bolts.

10. Insert two gaskets, one on each side of Orifice ML 0224 at point "G," and insert one gasket at point "F." Then assemble pipe PT. #3 and make up bolts.

11. Insert gasket at point "D," assemble ML 1023 (VPR2 valve) and make up bolts.

12. Fit pipe PT. #4 to pipe PT. #3 and VPR2 valve at point "H." Seal weld slip-flange PT. #6 to pipe PT. #4, and then seal weld pipe PT. #4 to pipe PT. #3.

Note: After seal welding pipe PT. #4 to pipe PT. #3, remove bolts from flanges at points "C," "G" and "H." Remove pipe PT. #3, clean, and then reassemble.

13. Fit and assemble PT. #33 tubing and PTS. #32 and #5010 running from the top of VPR2 valve to pipe PT. #5.

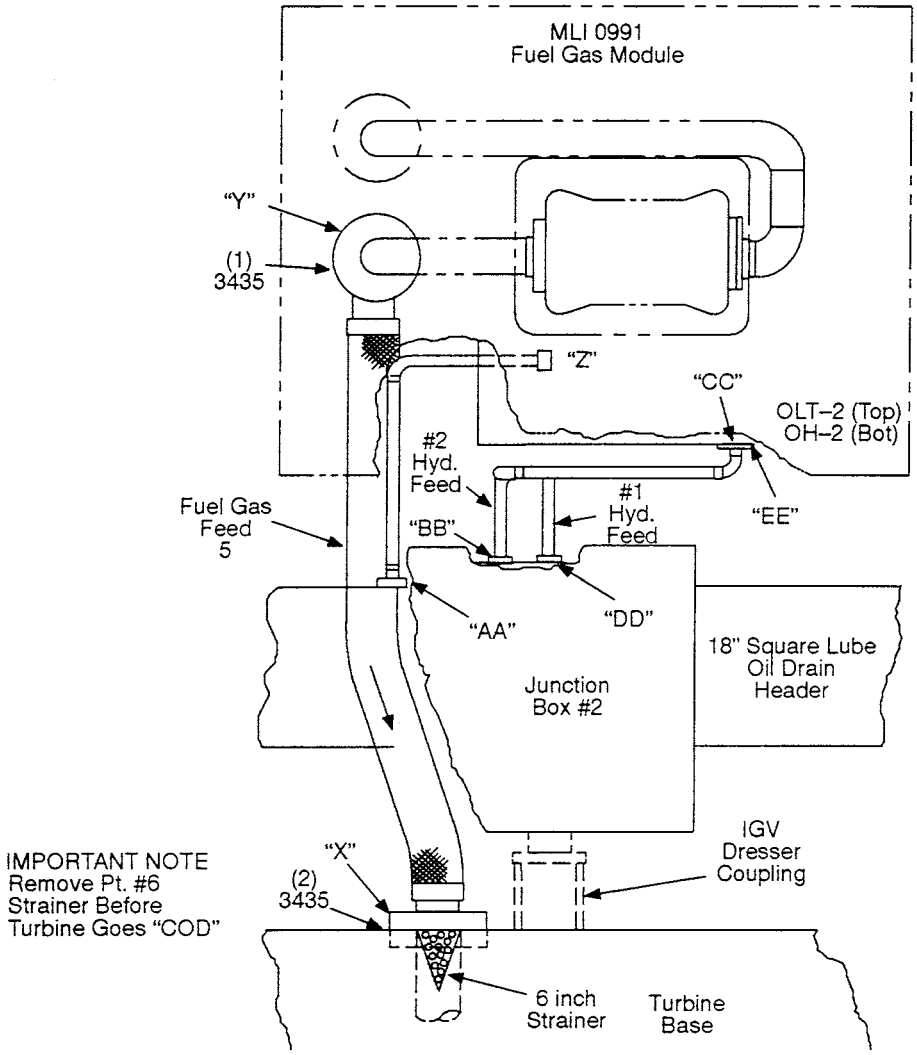
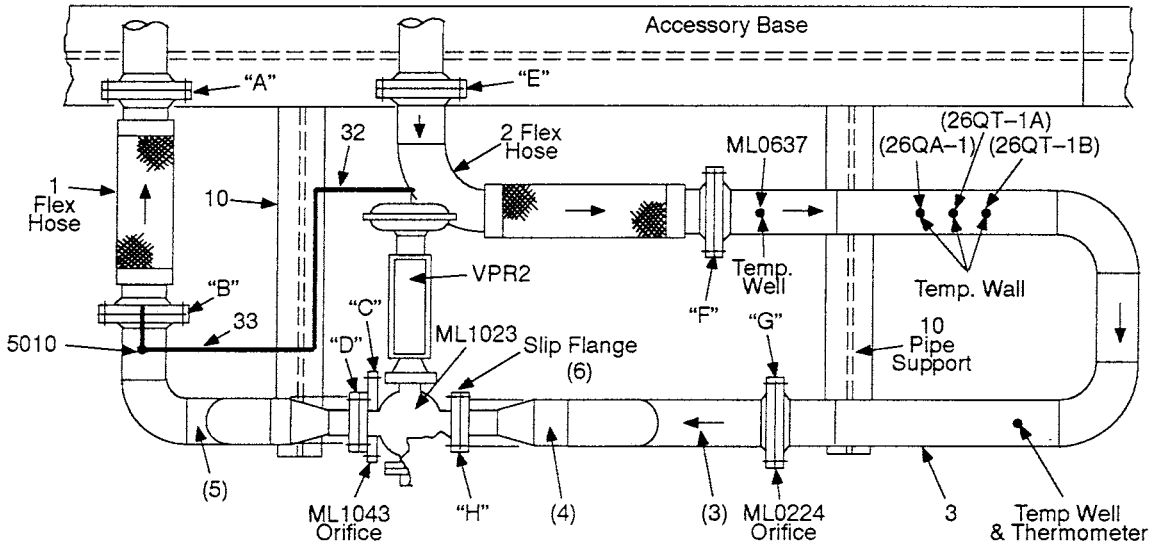


Figure 13. Field Assembly of Hydraulic and Fuel Gas Piping from Gas Module to Turbine



NOTE: Parts (3), (4), (5), (6) Found on Drawing 213C1284G001

NOTE: Parts 1, 2, 3, 10, 0224, 1043, 1023, 32, 33, 5010 Found on Drawing 103E8107G001

Figure 14. Assembly of Turbine Interconnecting Lube Oil Header